CLASS 8 · CBSE · GANITA PRAKASH PART 2 · SHORT NOTES

# Chapter 1: Fractions in Disguise

Topic-wise short notes with solved examples

Percentages are fractions with denominator 100. This chapter covers converting between fractions, decimals and percentages, finding a percentage of a quantity, percentage change, profit and loss, discount, tax, and growth with and without compounding.

## 1 Per cent

- **Per cent** means "out of a hundred". $x\% = \frac{x}{100}$.
- $25\% = \frac{25}{100} = \frac{1}{4} = 0.25$.
- **Fraction to percentage**: multiply by 100. $\frac{3}{4} \times 100 = 75$, so $\frac{3}{4} = 75\%$.
- **Percentage to fraction**: write over 100 and simplify. $24\% = \frac{24}{100} = \frac{6}{25}$.
- **Percentage to decimal**: divide by 100. $43\% = 0.43$, $5\% = 0.05$, $120\% = 1.2$.
- Percentages make comparison easy because every quantity is put on the same scale of 100.

| Per cent | 50% | 25% | 75% | 20% | 10% | 5% | 1% | $33\frac{1}{3}\%$ |
|---|---|---|---|---|---|---|---|---|
| Fraction | $\frac{1}{2}$ | $\frac{1}{4}$ | $\frac{3}{4}$ | $\frac{1}{5}$ | $\frac{1}{10}$ | $\frac{1}{20}$ | $\frac{1}{100}$ | $\frac{1}{3}$ |
| Decimal | 0.5 | 0.25 | 0.75 | 0.2 | 0.1 | 0.05 | 0.01 | 0.333... |

**Example 1.1** Express $\frac{2}{5}$, $\frac{72}{150}$ and $\frac{1}{3}$ as percentages.

SOLUTION

$\frac{2}{5} \times 100 = 40$; $\frac{72}{150} \times 100 = 48$; $\frac{1}{3} \times 100 = 33.33...$

**40%, 48% and $33\frac{1}{3}\%$**

**Example 1.2** 15 of the 80 students in a school walk to school. What percentage is this?

SOLUTION

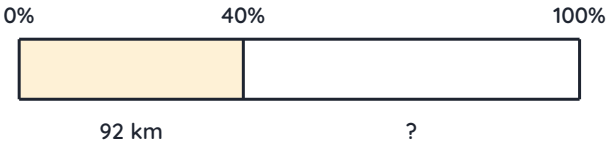
$\frac{15}{80} \times 100$

**18.75%**

## 2 Percentage of a quantity

- $y\%$ of a quantity $z$ is $\frac{y}{100} \times z$.

- **Mental shortcuts**: 10% is one-tenth; 5% is half of 10%; 20% is double 10%; 25% is a quarter; 15% $=$ 10% $+$ 5%; 75% $=$ 50% $+$ 25%; 90% $=$ 100% $-$ 10%.
- $x\%$ of $y$ equals $y\%$ of $x$, since both are $\frac{xy}{100}$. So 12% of 25 $=$ 25% of 12 $=3$.
- Estimate first. It catches mistakes.




Bar model: 40% of the journey is 92 km

**Example 2.1** Find 80% of 75 in two ways.

SOLUTION

As a fraction: $\frac{4}{5} \times 75 = 60$. As a decimal: $0.8 \times 75 = 60$.

**60**

**Example 2.2** A cyclist has covered 92 km, which is 40% of a journey. How far is left?

SOLUTION

40% is 92 km, so 20% is 46 km and 60% is $92 + 46$.

**138 km (the whole journey is 230 km)**

**Example 2.3** 40 is 80% of what number?

SOLUTION

$\frac{80}{100} \times n = 40$, so $n = 40 \times \frac{100}{80}$

**50**

**Example 2.4** Millet and water are mixed in the ratio $2:7$. What percentage of the mixture is millet? How much millet is in 450 mL?

SOLUTION

Millet is $\frac{2}{9}$ of the mixture: $\frac{2}{9} \times 100 \approx 22.2\%$.

$\frac{2}{9} \times 450$

**About 22.2%; 100 mL**

## 3 Percentages greater than 100

- A percentage above 100 means **more than the whole** used for comparison. 120% of a target is 1.2 times the target; 250% is 2.5 times.

**Example 3.1** A shop aims for sales of ₹5000 a day. It sells goods worth ₹2000 on Monday and ₹6000 on Tuesday. What percentage of the target is each?

SOLUTION

$\frac{2000}{5000} \times 100 = 40$ and $\frac{6000}{5000} \times 100 = 120$

**40% and 120%**

**Example 3.2** A farmer harvested 260 kg last year and 650 kg this year. What percentage of last year's harvest is this year's?

SOLUTION

$\frac{650}{260} \times 100$

**250%**

## 4 Comparing with percentages

- Marks out of different totals, or ingredients in packs of different sizes, are compared by converting to percentages.
- A percentage compares a part with **its own whole**. 35% of a small quantity can be less than 25% of a larger one.

**Example 4.1** Eesha scores 42 out of 50 in English and 70 out of 80 in Science. In which did she do better?

SOLUTION

English: $\frac{42}{50} \times 100 = 84\%$. Science: $\frac{70}{80} \times 100 = 87.5\%$.

**Science**

**Example 4.2** Biscuit A is 25% sugar and biscuit B is 35% sugar. One person eats 120 g of A and another eats 80 g of B. Who eats more sugar?

SOLUTION

A: $0.25 \times 120 = 30$ g. B: $0.35 \times 80 = 28$ g.

**The person eating A**

## 5 Percentage increase and decrease

$$\text{Percentage change} = \frac{\text{amount of change}}{\text{original amount}} \times 100$$

- An increase of $r\%$ multiplies the quantity by $\left(1 + \frac{r}{100}\right)$. A decrease of $r\%$ multiplies it by $\left(1 - \frac{r}{100}\right)$.

- "Increased by 65%" and "is 165% of the earlier value" say the same thing.
- "Fallen by 85%" means the new value is 15% of the old one.

**Example 5.1** The price of tomatoes rose from ₹30 to ₹42 per kg. Find the percentage increase.

SOLUTION

Increase $= 12. \dfrac{12}{30} \times 100$

**40%**

**Example 5.2** The audience in a hall fell from 160 to 100. Find the percentage decrease.

SOLUTION

$\dfrac{60}{160} \times 100$

**37.5%**

**Example 5.3** A number increased by 20% becomes 90. Find the number.

SOLUTION

$1.2 \times n = 90$

**75**

## 6 Profit and loss

- **Cost price (CP)**: what the seller paid. **Selling price (SP)**: what the buyer pays. **Marked price (MP)**: the price quoted before any discount.
- Profit $=$ SP $-$ CP when SP is greater. Loss $=$ CP $-$ SP when CP is greater.
- Profit and loss percentages are taken **on the cost price** unless stated otherwise.

$$\text{Profit}\% = \frac{\text{Profit}}{\text{CP}} \times 100, \qquad \text{Loss}\% = \frac{\text{Loss}}{\text{CP}} \times 100$$

- SP $=$ CP $\times \left(1 + \dfrac{\text{profit}\%}{100}\right)$ or CP $\times \left(1 - \dfrac{\text{loss}\%}{100}\right)$.
- **Gross profit** $=$ sales $-$ cost of the goods. **Net profit** $=$ gross profit $-$ other expenses.

**Example 6.1** A sweater bought for ₹300 is sold for ₹430. Find the profit percentage.

SOLUTION

Profit $= 130. \dfrac{130}{300} \times 100$

**About 43.3%**

**Example 6.2** Notebooks are bought at ₹36 each and sold at a profit of 20%. Find the selling price.

SOLUTION

$36 \times 1.2$

**₹43.20**

**Example 6.3** A box of crayons is sold for ₹50 at a profit of 25%. Find the cost price.

SOLUTION

CP $\times 1.25 = 50$

**₹40**

**Example 6.4** A vase bought for ₹2650 is sold at a loss of 18%. Find the selling price.

SOLUTION

$0.82 \times 2650$

**₹2173**

**Example 6.5** The selling price of 3 pencils equals the cost price of 5 pencils. Find the profit percentage.

SOLUTION

Let the CP of one pencil be ₹1. Then SP of 3 pencils $= 5$, so SP of one $= \frac{5}{3}$.

Profit on one $= \frac{2}{3}$, which is $\frac{2}{3} \times 100$ per cent of the cost.

**$66\frac{2}{3}$% profit**

## 7 Discount and tax

- **Discount** is a reduction on the marked price. Discount $=$ discount% of MP, and SP $=$ MP $-$ discount.
- **Successive discounts** such as "30% + 20%" are applied one after the other, so they are **not** the same as 50%. The price is multiplied by $0.7 \times 0.8 = 0.56$, a total discount of 44%.
- **Tax** such as GST is a percentage added to the price: final price $=$ price $\times \left(1 + \frac{\text{tax}\%}{100}\right)$.

**Example 7.1** A cooker with a marked price of ₹1800 is sold at a 35% discount. Find the selling price. If it cost the shop ₹900, find the profit percentage.

SOLUTION

SP $= 0.65 \times 1800 = 1170$. Profit $= 270$, and $\frac{270}{900} \times 100 = 30$.

**₹1170; 30% profit**

**Example 7.2** A car is bought for ₹4,40,000 after a 12% discount. Find the original price.

SOLUTION

$0.88\times$ MP $= 440000$

**₹5,00,000**

**Example 7.3** A phone costs ₹8250 and GST of 18% is added. Find the final price.

SOLUTION

$8250 \times 1.18$

**₹9735**

**Example 7.4** A shopkeeper marks goods 50% above cost and then gives a 50% discount. Does he break even?

SOLUTION

SP $= 1.5 \times 0.5\times$ CP $= 0.75\times$ CP

**No. He loses 25%. To break even, the discount must be $33\frac{1}{3}$%.**

## 8 Interest without compounding

- Money deposited or borrowed is the **principal** $p$. The extra money paid for its use is **interest**. The **rate** $r$ is a percentage per year (per annum, p.a.).
- If the interest is paid out each year, the principal stays the same and the interest each year is the same. After $t$ years:

$$\text{Interest} = p \times r \times t, \qquad \text{Amount} = p(1 + rt)$$

- Here $r$ is written as a decimal: 10% is 0.1.

**Example 8.1** ₹6000 is deposited at 10% p.a. for 3 years without compounding. Find the total received.

SOLUTION

Interest each year $= 600$. Total interest $= 1800$.

**₹7800**

## 9 Compounding

- If each year's interest is **added to the principal**, the next year's interest is calculated on a larger amount. This is **compounding**. Each year the amount is multiplied by $(1 + r)$.

$$\text{Amount} = p(1 + r)^t$$

| Year | Start | Interest at 10% | End |
|---|---|---|---|
| 1 | 6000 | 600 | 6600 |
| 2 | 6600 | 660 | 7260 |
| 3 | 7260 | 726 | 7986 |

- With compounding ₹6000 becomes ₹7986, a gain of 33.1%. Without it the gain is 30%.
- Growth without compounding is **linear** (adds the same amount each time). Growth with compounding is **exponential** (multiplies by the same factor each time).
- The same formula describes populations and other quantities that grow by a fixed percentage.

**Example 9.1** ₹20,000 is deposited at 10% p.a. for 2 years. Compare the amounts with and without compounding.

SOLUTION

Without: $20000(1 + 0.1 \times 2) = 24000$. With: $20000 \times 1.1^2 = 24200$.

**Compounding gives ₹200 more**

**Example 9.2** A city of 1.5 crore people grows by 3% every year. Find the expected population after 2 years.

SOLUTION

$1.5 \times 1.03^2 = 1.5 \times 1.0609$

**About 1.59 crore**

**Example 9.3** A fare rises by 3% one year and 4% the next. Find the overall percentage increase.

SOLUTION

$1.03 \times 1.04 = 1.0712$

**7.12%**

## 10 Depreciation and decline

- Many things lose value with age and use. This fall in value is **depreciation**. A fall of $r$ each period multiplies the value by $(1 - r)$.

$$\text{Value after } t \text{ periods} = p(1 - r)^t$$

**Example 10.1** A TV bought for ₹21,000 loses 5% of its value in a year. Find its value after one year.

SOLUTION

$0.95 \times 21000$

**₹19,950**

**Example 10.2** A village of 1250 people shrinks by 10% every decade. Estimate the population after 3 decades.

SOLUTION

$1250 \times 0.9^3 = 1250 \times 0.729$

**About 911**

## 11 Common mistakes

- Adding successive percentages: +10% then +10% is +21%, and 30% then 20% off is 44% off.
- Thinking +50% followed by $-50\%$ returns to the start. It leaves 75%.
- Using the selling price as the base for profit% or loss%. The base is the cost price.
- Using the new value as the base for a percentage change. The base is the original value.
- Finding the original price by adding the discount percentage to the sale price. Divide instead: MP $=$ SP $\div \left(1 - \frac{d}{100}\right)$.
- Comparing percentages of different wholes as if they were actual amounts.

MathSetu